Youssef Alikhani

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Research interests: Deep learning for time-series and sensor data (real-time signal processing, human activity recognition), computer vision, and real-time interactive systems.

Education

M.Sc. in Computer Science (Artificial Intelligence) Sep 2024 – expected Jun 2027
University of Tehran *Tehran, Iran*
Thesis: *Offline Handwritten Text Modeling using VLM & Diffusion Models* | Advisor: Dr. Bagher BabaAli | GPA: 15.56/20

B.Sc. in Computer Science Sep 2020 – Jun 2024
University of Mohaghegh Ardabili *Ardabil, Iran*
Supervisor: Dr. Mohamed Reza Hooshmand Asl | GPA: 18.29/20

Research Experience

Research Assistant Apr 2025 – Present
University of Tehran *Tehran, Iran*
Supervisor: Dr. Bagher BabaAli

- Researching offline handwritten text modeling with vision-language and diffusion models (M.Sc. thesis).
- Working with smartphone sensor data for locomotion and transportation mode recognition.
- Member of the team that won 1st place in the Sussex-Huawei Locomotion (SHL) Challenge 2025 (Task 1) with an ensemble of time-series foundation models; co-authored the paper at the HASCA Workshop, UbiComp/ISWC 2025.

Publications

[1] Mohammad Foad Abdi, **Yousef Alikhani**, Mohammad Mahdi Azizi, Bagher BabaAli, Mehdi Mohebbi, Arash Nasr Esfahani. "Ensemble of foundation models for sensor-based locomotion and transportation mode recognition." *Companion of the 2025 ACM International Joint Conference on Pervasive and Ubiquitous Computing (UbiComp/ISWC '25), pp. 949-955*, 2025. [link]

Teaching Experience

Teaching Assistant Sep 2022 – Jun 2023
University of Mohaghegh Ardabili *Ardabil, Iran*

- Computer Programming.
- Deep Learning.

Professional Experience

C++ Desktop Developer and Deep Learning Researcher Mar 2025 – Present
DFP *Tehran, Iran*

- Develop real-time Qt desktop applications in C++ for acquiring, processing, and visualizing sensor signals.
- Built a multi-threaded data pipeline with Taskflow and ZeroMQ (cppzmq) for low-latency processing and inter-process communication.
- Developed C++ simulators that generate synthetic signal data for signal-processing and deep learning tasks.
- Develop deep learning models for processing the acquired and simulated signal data.

Video Surveillance Engineer Mar 2024 – Oct 2024
Security systems company (banking sector) *Mashhad, Iran*

- Worked with IP camera and NVR-based video surveillance systems for bank security.

Honors & Awards

1st Place, Sussex-Huawei Locomotion (SHL) Challenge 2025 (Task 1), team member, Oct 2025
University of Sussex & Huawei, HASCA Workshop at UbiComp/ISWC 2025. International competition on locomotion and transportation mode recognition from smartphone sensor data.

Projects

IMS600 — Standalone C++/Qt desktop application for controlling and monitoring an Ion Mobility Spectrometry (IMS) instrument: live signal visualization, spectrum detection, calibration, and device control (high voltage, pulse generator, pump, heater, valves).

Skills

Programming: C++, Python

Machine learning: PyTorch, NumPy, OpenCV

C++ libraries: Qt, Boost, Armadillo, Taskflow, ZeroMQ (cppzmq), FreeRTOS, GoogleTest

Graphics and games: Godot, raylib, Dear ImGui

Tools: Git, Linux, CMake

Languages

Languages: Persian (Native), English